

DEMO 05 • IMAGING CONTEXT

Catch the Bad Study Before the Radiologist Does

A read-only DICOM context interpreter that raises a QC advisory on missing metadata and a routing advisory on a complete study — without ever touching a pixel.

READ-ONLY Reads metadata — no writeback

AUTHOR Rakovskyi Dmytro • VERSION Full-run demo 05 • DATE 2026 • REPOSITORY github.com/rakovpublic/jneopallium

A study with missing metadata wastes a radiologist's time twice: once when it lands in the wrong worklist, and again when it has to be sent back. This demo reads only the metadata around an imaging study — modality, body part, series count, study age — and sorts the good from the incomplete before a human ever opens it. It diagnoses nothing in the image and writes nothing back; it is pure, careful triage of context.

What it is

Demo 05 models a DICOM context bridge. Its safety ceiling is **READ-ONLY**: it consumes and annotates imaging metadata but performs no pixel diagnosis and no writeback to the imaging system. The network is just three layers (sized 3·2·2) — the smallest stack that turns metadata into a routing or quality-control advisory.

How it is wired

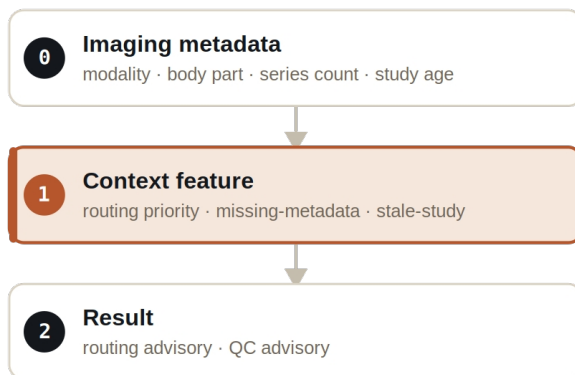


Figure Three layers. The context-feature layer flags missing or stale metadata; the result layer advises routing or QC.

Signals are typed and narrow: `DicomMetadataSignal` carries the study metadata, `ImageQcSignal` carries quality-control findings, and `RoutingAdvisorySignal` carries the routing recommendation. Nothing in the vocabulary can express a diagnosis or a write.

What it does

- A complete study is given a `ROUTING_ADVISORY` — sent to the right worklist with the right priority.
- A study with missing metadata is given a `QC_ADVISORY` so it can be corrected before it reaches a radiologist.
- No result reads pixel data, infers a clinical finding, or writes anything back to the source.

NOTE

The boundary is deliberately strict: the demo is a metadata triage assistant, not an image reader. The test proves no command or write result exists at all.

The evidence

VERIFICATION EVIDENCE	
Ticks	60 / 60
Complete study	<code>ROUTING_ADVISORY</code>
Missing metadata	<code>QC_ADVISORY</code>
Writeback / pixel diagnosis	none — enforced
Output mode	READ-ONLY

How to run it

1 Build

Build the worker and the demo model JAR.

```
mvn -q -pl worker -am package
```

2 Run via the framework entry point

Entry loads the model JAR and the demo context through the real local path.

```
java -cp demo-model.jar \  
com.rakovpublic.jneuropallium.worker.application.Entry \  
local file:///path/to/demo-model.jar \  
com.rakovpublic.jneuropallium.worker.demo.fullrun.runtime.DemoJsonContext \  
context.json
```

3 Inspect

Read the routing and QC advisories.

```
ls target/jneopallium-fullrun-demos/demo-05-dicom-context/
```

Why READ-ONLY

Pixel-level interpretation is a regulated diagnostic act and writeback to a PACS is an operational risk; this demo deliberately does neither. The read-only ceiling keeps it a context assistant. To connect it to a live archive you replace the mock metadata edge with a real DICOM source while preserving the typed signal contract and the read-only ceiling — and the guarantee that it never writes or diagnoses comes along unchanged.